Immanuel Eben Haezer Joseph Aletheia

Software Engineer: Web, Applications, and Embedded Systems | Final-Year Electrical Engineering Student

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SUMMARY

Software engineer working across full-stack web and application development in Python, JavaScript, and C/C++. Currently building an asset-management platform for PT Kereta Api Indonesia on a FastAPI backend, with WebSocket live updates, JWT authentication, and response compression. Earlier work for the same client shipped a computer-vision system into use, alongside a machine-learning product with an Android front end. Also writes embedded firmware in C/C++ for ESP32-class boards, covering a nestest-verified NES emulator core, PID closed-loop control, and multi-sensor instrumentation. Final-year Electrical Engineering student, GPA 3.77/4.00 with 125 of 145 credits completed. Looking for a software engineering role in web or application development where the hardware background counts as a bonus rather than the job itself.

SKILLS

Languages: Python, JavaScript, C, C++, SQL, HTML/CSS, MATLAB
Web & Backend Development: FastAPI, REST API design and integration, WebSocket, JWT authentication, rate limiting, response-compression middleware, relational database design, Firebase Realtime Database, Vercel
Application & Machine Learning: Android Studio, LVGL touchscreen UI, computer vision (OpenCV), machine learning (TensorFlow/TFLite Micro), image segmentation
Embedded Systems & IoT: ESP32/ESP32-S3, Arduino/PlatformIO, firmware in C/C++, MQTT, HTTP, UART, I2C, sensor and actuator integration (load cells, IMU, gas sensors), PID closed-loop control, LVGL and TFT touchscreen HMI, TFLite Micro on-device inference
Tools & Platforms: Git, GitHub, Linux, Visual Studio Code, PlatformIO, KiCad, Proteus, MATLAB/Simulink
Spoken Languages: Indonesian (native), English (professional working, EPRT 542), Mandarin (elementary), Japanese (elementary)

EXPERIENCE

Software Engineering Intern (Fullstack) — Jul 2026 - Present

PT KAI (Kereta Api Indonesia), Track & Bridge Division, Bandung, Indonesia

Tech: Python, FastAPI, SQL, WebSocket, JWT authentication, JavaScript

- Shipped RAMCES v1.0.1, a full-stack asset-management platform for PT KAI's Track & Bridge division, from the data model through to the screen: 122 REST and WebSocket endpoints over a 22-table PostgreSQL schema, now running against the real fleet, 104 equipment types across 273 locations in 16 regions nationwide. Around 55,000 lines of Python and JavaScript, 153 commits, in a team of three.
- Wrote the FastAPI backend's WebSocket live-update layer, JWT authentication across five region-scoped roles, rate limiting, and the response-compression middleware that cuts a 470 KB JS and 383 KB HTML payload by roughly 8:1.
- Refactored the backend from a single 5,900-line file into a package of eleven domain modules, holding a no-cyclic-imports rule that the original layout could not express.
- Repository: github.com/EintsWaveX/RAMCES_LightMachinery

Research Assistant, Programming Division — Feb 2024 - Present

LumiaTech Innovations Research Laboratory, Telkom University, Bandung, Indonesia

Tech: Python, C++, OpenCV, TensorFlow, image segmentation

- Shipped a railway-track anomaly detection system for PT Kereta Api Indonesia, using OpenCV image segmentation on trackside video to flag defects along the rail corridor. Completed and accepted, in a team of three.
- Built a multi-view reconstruction module that recovers several usable viewpoints from a single monocular fisheye camera.

PROJECTS

SkyNES: a NES Emulator Core Built From Scratch — Jun 2026 - Present

Tech: C++, 6502 CPU emulation, PPU, ESP32-S3, PlatformIO

- Wrote a complete NES emulator core in C++ single-handedly, covering the 6502 CPU, the PPU, and cartridge/mapper handling. Verified the CPU byte for byte against the nestest golden reference trace across all 8,990 instructions, and now porting it to a custom ESP32-S3 handheld with a 4.0″ TFT touchscreen.
- Hardware bring-up (display wiring) is still in progress; the core logic runs correctly end to end on the desktop test harness today.
- Repository: github.com/EintsWaveX/SkyNES

Smart Rack Monitoring System
Jan 2026 - Jun 2026

Tech: ESP32, ESP32-S3, LVGL, HX711 load cells, MPU6050, UART, C++

- Wrote both firmware halves of a dual-ESP32 industrial rack-monitoring system for a Capstone Design project (5 credits): an ESP32-S3 driving a 480×320 LVGL touchscreen dashboard, and a sensor node reading four HX711 load cells, one per rack footing, plus an MPU6050 IMU over UART.
- Measures loads above 200 kg across all four corners, accurate to within 1 kg per load cell and roughly 4 kg across the whole rack, shown as a live circular gauge with per-corner readouts, and warns of overload through colour, LED, and buzzer.
- Repository: github.com/EintsWaveX/SmartRackMonitoring

Community Air Quality Monitoring System, Sindangkerta
Apr 2026 - May 2026

Tech: JavaScript, Vercel, Firebase Realtime Database, ESP32, gas sensors

- Built and deployed the public web dashboard on Vercel, which streams readings from the field station through Firebase Realtime Database and renders them for the community. The dashboard is still publicly deployed.
- Delivered as a completed community-service (pengmas) initiative built around a five-sensor station measuring PM2.5/PM10, O_3, H_2S, CO, and NO_2.
- Live dashboard: web-pengmas.vercel.app

Intelligent Waste Sorting System
Nov 2025 - Dec 2025

Tech: Python, TensorFlow, TFLite Micro, ESP32-CAM, Firebase, Android Studio

- Built a waste-sorting system that runs end to end. A custom lightweight CNN (TensorFlow, INT8-quantized to TFLite Micro) on an ESP32-CAM classifies plastic, paper, and metal from a live camera feed in 10 to 150 ms, at 85% to 89% accuracy across four input resolutions, reaching 91% on plastic and 96% on paper. A servo-driven rotating platform then sorts the item into its bin. Owned the ML, the firmware, and the app, in a team of three.
- Synced live device state to Firebase Realtime Database at 50 to 150 ms latency for a custom Android monitoring app.
- Repositories: TrainingWorkspace • IoTxAI • ES

PID Temperature and Humidity Control Prototype
Apr 2025 - Jun 2025

Tech: C++, PID control, DHT22, TFT touchscreen, Arduino

- Built and tuned a closed-loop PID temperature controller for a humidified enclosure: a DHT22 feeds a reverse-acting PID loop (Kp 100, Ki 0.1, Kd 5.0) that modulates a fan blower on 25 kHz PWM against a water pump on 1 kHz PWM, holding setpoints across a 24 to 32°C range.
- Built the 3.2″ TFT touch interface, including a live rolling graph of temperature, humidity, and PID output over a 210-sample history. Main contributor on a team project, set as joint coursework for Sensors & Actuators and Basic Control Systems.
- Repository: github.com/EintsWaveX/EL4705_09_TugasBesarSNAxSKD

TEACHING & ACADEMIC LEADERSHIP

Practicum Assistant & Division Coordinator
Sep 2024 - Present

Basic Computing Laboratory (C) and IRIS Laboratory (IoT), Telkom University, Bandung, Indonesia

- Write C and IoT programming problems, test cases, and grading rubrics each semester. Lead the Assignment and Task Committee, and ran the academic track of DLOR 2025 assistant recruitment.

Lecturer's Assistant, Digital Engineering and Calculus I
May 2025 - Jan 2026

Telkom University, Bandung, Indonesia

- Taught and assessed two undergraduate courses, designing quizzes, examinations, and projects.

EDUCATION

Bachelor of Engineering, Electrical Engineering
Expected Sep 2027

Telkom University, Bandung, Indonesia

- GPA 3.77/4.00. Relevant coursework: Digital and Microcomputer Engineering, Artificial Intelligence and Machine Learning, Embedded Systems, Control Systems, Internet of Things.